

A SYNERGISTIC PHYTO-LIPOSOMAL TOPICAL COSMETIC COMPOSITION COMPRISING HARIDRA AND KUMKUMADI ACTIVES FOR ENHANCED DERMAL ABSORPTION AND SKIN RADIANCE

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REGULATORY CATEGORY:	Ayurvedic Cosmetic (Drugs & Cosmetics Act 1940, Section 3(aaa))
IPC / CPC CLASSIFICATION:	A61K 8/9789 (Cosmetics from Plant extracts), A61Q 19/02, A61Q 19/08

1. FIELD OF THE INVENTION

The present invention relates to the technical field of topical cosmetic compositions and personal care aesthetics. More particularly, the invention relates to a non-staining, stable sub-micron phyto-liposomal cosmetic emulsion incorporating standardized extracts of Haridra (*Curcuma longa* rhizome) and Kumkumadi botanical actives (*Crocus sativus* stigma and *Pterocarpus santalinus*) in a phospholipid vehicle for topical application to skin. The formulation is strictly intended for cleansing, beautifying, and enhancing skin radiance without claiming therapeutic disease mitigation under Section 3(aaa) of the Drugs and Cosmetics Act, 1940.

2. BACKGROUND & STATUTORY PATENTABILITY OVERCOMING SECTION 3(p)

Classical Ayurvedic preparations such as Kumkumadi Tailam and Haridra Lepa are extensively documented in ancient treatises, including Ashtanga Hridaya and Bhaishajya Ratnavali (Kshudraroga Rogadhikara), and cataloged in the Traditional Knowledge Digital Library (TKDL). Accordingly, crude preparations of turmeric and saffron per se are unpatentable in India under Section 3(p) of the Patents Act, 1970 as traditional knowledge, and under Section 3(e) as mere admixtures.

However, conventional cosmetic use of turmeric suffers from three severe commercial handicaps: (i) intense yellow staining on human epidermis, (ii) rapid photodegradation of active curcuminoids upon sunlight exposure, and (iii) poor stratum corneum penetration (> 92% remains on skin surface). The present invention solves these technical problems through a proprietary sub-micron phospholipid lipid bilayer encapsulation exhibiting an average vesicle size of 124.8 nm and a polydispersity index of 0.14. This technical intervention overcomes the Section 3(p) traditional knowledge bar by creating an artificial delivery matrix not known in classical Samhitas.

3. DETAILED COMPOSITION & SYNERGISTIC EMBODIMENTS

A preferred embodiment of the topical cosmetic cream comprises:

- Standardized *Curcuma longa* extract (enriched in Tetrahydrocurcumin 95%): 1.5 wt%
- *Crocus sativus* (Kashmir Saffron) stigma extract (Crocin 12%): 0.5 wt%
- Hydrogenated Soy Phosphatidylcholine (Phospholipon 90H): 4.0 wt%
- Plant-derived Squalane carrier: 6.0 wt%
- Jojoba Seed Oil (*Simmondsia chinensis*): 3.5 wt%
- Cetearyl Olivat and Sorbitan Olivat emulsifier base: 3.0 wt%
- Purified Rose Water (*Rosa damascena* distillate): Q.S. to 100 wt%

Spectrophotometric skin permeation assays (Franz diffusion cell) established a 6.2-fold increase in dermal retention over traditional tailam, while primary skin irritation testing in accordance with Bureau of Indian Standards (BIS IS 4707) confirmed a zero irritation score (PII = 0.0), making it completely compliant with Indian cosmetic regulatory standards.

4. PATENT CLAIMS (WE CLAIM)

1. A synergistic, non-staining topical cosmetic phyto-liposomal composition comprising:
 - (a) a botanical active complex comprising 1.0 to 3.0 wt% of standardized *Curcuma longa* rhizome extract and 0.2 to 1.0 wt% of *Crocus sativus* stigma extract;
 - (b) a lipid vesicle bilayer matrix comprising 3.0 to 6.0 wt% of hydrogenated phosphatidylcholine; and
 - (c) a cosmetically acceptable carrier comprising 5.0 to 10.0 wt% plant squalane and rose distillate;wherein the botanical active complex is encapsulated within sub-micron liposomal vesicles having an average diameter of 110 nm to 140 nm and a polydispersity index of less than 0.18, thereby preventing epidermal yellow discoloration.
2. The cosmetic composition as claimed in claim 1, wherein the composition exhibits a synergistic tyrosinase inhibition Combination Index (CI) of 0.62, demonstrating statistical synergism under Section 3(e) of the Indian Patents Act, 1970.
3. The cosmetic composition as claimed in claim 1, wherein the formulation is formulated as a topical cosmetic cream, gel-cream, or serum conforming to BIS IS 4707 standards for skin aesthetics.
4. A process for preparing the phyto-liposomal cosmetic composition of claim 1, comprising the steps of:
 - (i) dissolving standardized *Curcuma longa* and *Crocus sativus* actives in organic ethanol with phosphatidylcholine at 45 deg C;
 - (ii) performing rotary thin-film hydration with aqueous rose distillate;
 - (iii) subjecting the hydrated dispersion to high-pressure microfluidization at 800 bar to yield uniform vesicles; and
 - (iv) homogenizing the vesicles into a cosmetically acceptable cream base under ambient sterile conditions.
5. The process as claimed in claim 4, wherein the biological resources originate from Maharashtra and Jammu & Kashmir, complying with geographical origin disclosure mandates under Article 3 of the WIPO GRATK Treaty, 2024 and Section 7 of the Biological Diversity Act, 2002.

5. STATUTORY REGULATORY & LICENSING COMPLIANCE

- Governing Authority: State AYUSH Licensing Authority / FDA Maharashtra (Form 32-A Cosmetic License).
- Safety Standard: Bureau of Indian Standards (BIS IS 4707 Part 1 & Part 2).
- ABS Compliance: Section 7 intimation to Maharashtra State Biodiversity Board (MSBB) prior to commercial manufacture.
- Statutory Restriction: Product packaging shall not bear any claim regarding therapeutic cure or treatment of disease as mandated under Rule 158-B and the Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954.